

**MBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE –
RAIGAD -402 103
Semester Examination – Dec - 2022**

Branch: Electronics and Telecommunication Engineering

Sem.:- V

Subject with Subject Code: - Digital Signal Processing (BTXC504) Marks: 60

Date:-

Time:- 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

(Marks)

- Q.1 A) Write the applications of digital signal processing. (04)
- B) Perform the following operations on signal $x(n) = \{1 \ 2 \ 4 \ 5\}$, $-1 \leq n \leq 2$ (04)
- a. $x(-n-1)$ b. $0.5x(n-2)$
- C) Find autocorrelation of $x(n) = \{1 \ 2 \ 1 \ 1\}$, $0 \leq n \leq 3$ (04)
- Q.2 A) Prove that real part of DTFT is even symmetric and imaginary part is odd symmetric. (04)
- B) Find the inverse Fourier transform of $H(e^{jw}) = \delta(w)$ $-\pi < w \leq \pi$ (04)
- C) State and prove Parseval's relation. (04)
- Q.3 A) Find the impulse response of the system described by the difference equation (06)
- $y(n] - 3y(n-1) - y(n-2) = x(n) + 2x(n-1)$ using Z-transform.
- B) Find Z-transform and plot the respective ROCs (06)
1. $0.5^n u(n) + 2^n u(-n-1)$
2. $-0.5^n u(-n-1) + 2^n u(-n-1)$

Q.4 A) State and prove properties of twiddle factor. (06)

B) Find the circular convolution of the given sequences using matrix multiplication method. $x_1(n) = \{1, 2, 3, 4\}$, $x_2(n) = \{1, 1, 2, 2\}$ (06)

Q.5 A) Compute the 8 point DFT of the following sequence using DIF-FFT algorithm. (06)
 $x_1(n) = \{1, 2, 1, 2, 1, 2, 1, 2\}$

B) Derive transfer function for high pass FIR digital filter and find out its cut-off frequency. Draw its pole zero plot. Plot its magnitude and phase response. (06)

Q.6 A) For $H(z) = 1 + 2z^{-1} - z^{-2} + 3z^{-3} + 3z^{-4} - z^{-5} + 2z^{-6} + z^{-7}$ (06)
Draw the linear phase FIR implementation.

B) Derive the equation for the order of the low pass Butterworth analog filter. (06)
